PLANTATION AND HORTICULTURE

Previous Year Question (2019-2023)

1. Fruits which continue to ripen even after harvest are called as

a. Non climacteric fruits
b. Climacteric fruits
c. Perishable fruits
d. None of the above

2. Olericulture is the scientific study of

a. Fruits
b. Forest plantation
c. Vegetables
d. Flowers

3. Pomology is a branch of horticulture which deals with the scientific study of

a. Fruits
b. Vegetables
c. Flower
d. None of the above

4. Floriculture is the term used for cultivation, harvesting and marketing of which of the following

a. Fruits
b. Forest trees
c. Vegetables
d. Flowers

5. Which of the following fruit or vegetable is a rich source of vitamin A essential source of retinol

a. Carrot
b. Brinjal
c. Avocado
d. None of the above

6. Fruits which are grown in cold region where temperature falls down below freezing point are called as

a. Subtropical fruits
b. Temperate fruits
c. Tropical fruit
d. None of the above

7. Geometric planning and arrangement of horticulture plants and flower for the purpose of beautification is called as

a. Landscape gardening
b. Forestry
c. Floriculture
d. None of the above

8. Vegetative propagation by the runners is a method used to propagate which fruit crop

a. Grapes
b. Guava
c. Strawberry
d. None of the above

9. Which planting material is used to grow potato in field

a. Tuber

b. stem

c. both

d. none of the above

10.Hard wood cuttings are generally used to propagate

a. Chrysanthemum

b. strawberry

c. guava

d. grapes

11.Guvava is generally commercially propagated by which method

a. air layering

b. soft wood cutting

c. stem budding

d. none of the above

12. which method of propagation is commonly practiced to propagated mango plants

a. soft wood cutting

b. hard wood cutting

c. inarching

d. none of the above

13. The process removes dead and dying branches and stubs, allowing room for new growth and protecting your property and passer-by from damage

a. training

b. pruning

c. propagation

d. none of the above

14. The transfer of pollen grain from the anther to the stigma of the same flower is which type of pollination

a. self-pollination

b. cross pollination

c. sexual reproduction

d. asexual reproduction

15. what is the scientific name of apple?

a. Malus domestica b. Pisum stivum c. Mangifera indica d. none of the above

16. The process of sorting out of fruit and vegetable on the basis of shape, size and colour

a. Sorting

B. Grading

C. Training

D. None Of The Above

17. Which of the following hormone is used as a rooting hormone in cutting during asexual propagation of plants.

a. auxin

b. ethylene

c. gibberellin

d. all of the above

18.Fruits and vegetables are generally dipped in water after harvesting with the main purpose of

a. Removal of field heat
b. For cleaning and removal of excess dirt
c. Both
d. None of the above

Answers

1.b	**2.c**	**3.a**	**4.d**	**5.a**	**6.b**	**7.a**	**8.c**	**9.a**	
10. d	**11.a**	**12.c**	**13.b**	**14.a**	**15.a**	**16.b**	**17.c**	**18.c**	

PLANTATION AND HORTICULTURE

The term Horticulture is derived from the **Latin words: "hortus" meaning garden** and **"cultura" meaning cultivation**. In ancient days the gardens **had protected enclosures with high walls or similar structures surrounding the houses**. The enclosed places were used to grow fruit, vegetables, flowers and ornamental plants. Therefore, in original sense "Horticulture refers to cultivation of garden plants within protected enclosures".

At present the horticulture may be defined as the science and technique of production, processing and merchandizing of fruits, vegetables, flowers, spices, plantations, medicinal and aromatic plants.

Importance Of Horticulture in The National Economy

India is the **2nd largest producer in the world**, in all, horticulture crops occupy an area of **23 million hectare and are said to annually yield 257 million tons of crops**. Horticulture sector not only impact the immediate life but play a very important role in the Indian economy by proving to be an **important source of income for the rural population**.

Branches of Horticulture

Horticulture is a wide field and **includes a great variety and diversity of crops**. The science of horticulture can be divided into several branches depending upon the crops it deals with.

The following are the branches of horticulture.

1. **Pomology** - It is derived from two words i.e. Pomum meaning fruit and Logos meaning discourse or study. Therefore, **pomology is study or cultivation of fruit crops** such as Mango Litchi, Citrus, Sapota, Guava, Grape, Banana, Pineapple, Apple, Pear, Peach, Plum and Cherry etc.

2. **Olericulture**: It is derived from two words i.e. Oleris meaning Potherb and Cultra meaning cultivation. Therefore, olericulture literally means **potherb cultivation of vegetables like Brinjal, Okra, Tomato, Capsicum, Peas, Beans, Cucurbits etc.**

3. **Floriculture:** It is derived from two words i.e. Florus meaning flower and Cultra meaning cultivation. Therefore, **floriculture means study of flower crops** such as Rose, Jasmine, Carnation, Aster, Marigold, Dahlia, Zinnia, Cosmos, Hibiscus, Balsam, Poinsettia, Hollyhock, Gerbera, and Gaillardia etc

4. **Landscape gardening**: It deals with the **planning and execution of ornamental gardens**, parks, landscape gardens etc.

5. **Plantation crops**: These **crops are cultivated in an extensive scale in large contiguous areas, owned and managed by an individual** or a company and whose produce is utilized only after processing. **Coffee, Tea, Rubber, Coconut, Cocoa etc. are some of the important plantation crops.**

6. **Spices and condiments:** This branch deals with the cultivation of crops whose produce is used mainly for seasoning and flavouring dishes.

   Spices: These are those plants whose products are **made use of as food adjuncts to add aroma and flavour**. For example, **Pepper, Cardamom, Clove, Cinnamon**, etc.

   Condiments: These are t**hose plants the products of which are made use of as food adjuncts to add taste only**. For example, **Turmeric, Ginger, Red chillies**, Onion, Garlic etc

7. **Medicinal and aromatic plants**: It deals with the cultivation of medicinal plants, which provide drugs and aromatic crops which yields aromatic (essential) oils.

   Medicinal plants: These **plants are rich in secondary metabolites and are potential sources of drugs**. The secondary metabolites include alkaloids, glycosides, coumarins, flavonoids and steroids etc. Important medicinal plants are Periwinkle, Opium, Menthi, Cinchona, Dioscorea Yam, Belladona, Senna, Sarpagandha, Ashwagandha, Tulasi etc.

<u>Aromatic plants</u>: These plants **possess essential oils in them**. The essential oils are the odoriferous steam volatile constituents of aromatic plants. **Lemon grass, Citronella, Palmrosa, Vetiver, Geranium, Davanam, Lavendor** etc. are some of the aromatic plants.

8. **<u>Post harvest technology</u>**: It deals with the **processing and preservation** of produce of horticulture crops.

Figure 1 – Pomology

Figure 2 – Olericulture

Figure 3 – Floriculture

Figure 4 – landscape gardening

Figure 5 – Plantation Crops

Figure 6 – Spices

Figure 7 – Condiments

Figure 8 – Medicinal Plants

Figure 9 – Aromatic Plants

Features of Horticulture

1. Horticultural produces are **mostly utilized in the fresh state and are highly perishable**.

2. Horticultural **crops need intensive cultivation** requiring a large input of capital, labour and technology per unit area.

3. are skilled and specific to horticultural crops. **Cultural operations like propagation, training, pruning and harvesting**

4. Horticultural produce are **rich sources of vitamins and minerals** and alkaloids.

5. **Aesthetic gratification** is an exclusive phenomenon to horticultural science.

Importance of Horticulture

Horticulture is important for the following considerations:

1. As a source of **variability in produce**.
2. As a **source of nutrients**, vitamins, minerals, flavour, aroma, alkaloids, oleoresins, fibre, etc.
3. As a **source of medicine**.
4. As an **economic proposition as they give higher returns** per unit area in terms of energy, money, job, etc.
5. **Employment generation 860-man days/annum** for fruit crops as against 143-man days/annum for cereal crops and the crops like grapes, banana and pineapple need 1000- 2500-man days per annum.
6. **Effective utilization of waste land** through cultivation of hardy fruits and medicinal plants.

7. As a substitute of **family income being component of home garden**.
8. As a **foreign exchange earner,** has higher share compare to agriculture crops.
9. As an **input for industry being amenable** to processing, especially fruit and vegetable preservation industry.
10. **Aesthetic consideration** and protection of environment.
11. **Religious significance**.

Functions Of Fruits and Vegetables in Human Nutrition

1. Vitamin-A-

It is essential for **normal growth, reproduction and maintenance of health** and vigour. It affords protection against cold and **influenza and prevents night blindness**. The deficiency of this vitamin results in cessation of growth in young children, night blindness, drying up of **tear glands in the eyes**, eruption of skin (Rashes on the skin) and brittleness of the teeth.

Sources:

Fruits-**Mango, Papaya, Dates, Jackfruit, Walnut etc**.

Vegetables-**Greens like palak, spinach**, Amaranthus, fenugreek, **carrot**, cabbage, lettuce, peas, tomato etc.

2. Vitamin B1 (Thiamine)

Tones the nervous system and helps in proper functioning of the digestive tract. Its deficiency **in human diet results in Beri-Beri, paralysis**, loss of sensitivity of skin, enlargement of heart, loss of appetite, loss of weight and fall in body temperature.

Sources:

Fruits-**Orange, pineapple, jack fruit**, cashew nut, walnut, dry apricot, almond, banana etc,

Vegetables-**Green chilli**, beans, onion, sweet potato, tomato (red), leaves of Colocasia.

3. Vitamin B2 (Riboflavin)

This vitamin is **required for body growth and health of the ski**n. The deficiency of this vitamin causes sore throat, anorexia cataract, and loss of appetite and body weight and also development of swollen nose.

Sources:

Fruits - Bael, papaya, litchi, banana, apricot, pomegranate, pear etc.

Vegetables - Cabbage, cauliflower, potato, peas and beans, meethi, lettuce, asparagus, green chillies, leafy vegetables etc,

4. Vitamin -C (Ascorbic Acid):

This vitamin promotes general health and **healthy gums, prevents scurvy disease** which is characterized by pain in the joints and swelling of limbs (rheumatism), bleeding of gums, tooth decay and keeps the blood vessels in good condition.

Sources:

Fruits: Amla, guava, ber, citrus, strawberry, pineapple etc.

Vegetables: Tomato, palak, menthi, cabbage, green chillies, spinach, potatoes, peas and beans and carrot etc,

5. Vitamin-D

This vitamin is **necessary for building up of bones**, preventing rickets and diseases of teeth.

Sources:

All green leafy vegetables are rich in this vitamin.

6. Vitamin-E

It has an important effect on the **generative functions and promotes fertility**.

Sources:

Green lettuce and other green vegetables, as well as almonds, cashew-nut, walnut etc.

7. Vitamin-K

This vitamin **prevents blood clotting**.

Sources: **All green leafy vegetables** and nuts are rich in this vitamin.

Classification Of Fruit Crops

Fruits are classified on various basis like part consumed, seeded or seedless, but here we only discuss about the fruit classification on the basis of climatic adaptations and on the basis of respiration.

A. Classification Of Fruits Crops on The Basis Of Climate Adaptabilities

Fruits

- **Tropical Fruit**
  ex- mango
- **Subtropical Fruit**
  ex-citrus fruit
- **Temprate Fruit**
  ex- apple

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1. **Tropical fruits: -** The fruits **require hot and humid climate** in Summer & mild winter.

   Example - Mango, Pineapple, Papaya, Cashew, Guava, Sapota, Custard Apple, Jackfruit, Coconut, Banana

2. **Subtropical fruits:** - The fruits **require hot and dry Climate** & Comparatively less winter.

   Example - Citrus Fruit, Karonda, Grape, Ber, Avocado, Fig, Bael, Litchi, Aonla, Jamun, Phlalsa, Pomegranate

3. **Temperate fruits: -** Fruit **require Cold winter where temperature fall below freezing** point. **They shed their leaves in winters** & go under dormancy.

Example-Apple, Almond, Pear, Peach, Plum, Strawberry, Walnut, Apricot, cherry, etc.

B. Classification On the Basis of Rate of Respiration: -

Fruits

Climatric fruits
ex Mango

Non-climatric fruits
ex-grapes

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1. Climacteric fruit: - Fruit continue to ripen after harvesting.

During ripening process, fruit emits ethylene (ripening hormone) along with increase in rate of **respiration & Co_2 production**.

After **harvesting, small dose of ethylene is used to induce ripening** process under controlled condition of temperature & humidity.

Example: - Mango, Sapota, Papaya, Fig, Peach, Pear, Plum, Apple, Banana, Guava

Figure 8 Climacteric & Non-Climacteric Fruits

2. Non climacteric fruit: - Fruit **which do not ripen after harvesting**.

They produce **very small amount of ethylene** and do not respond to ethylene treatment. There is **no characteristic increase in rate of respiration** or Co_2 production.

Example: - Citrus, Grape, Pineapple, Pomegranate, Litchi, Ber, Jamun, Cashew, Cherry, Watermelon

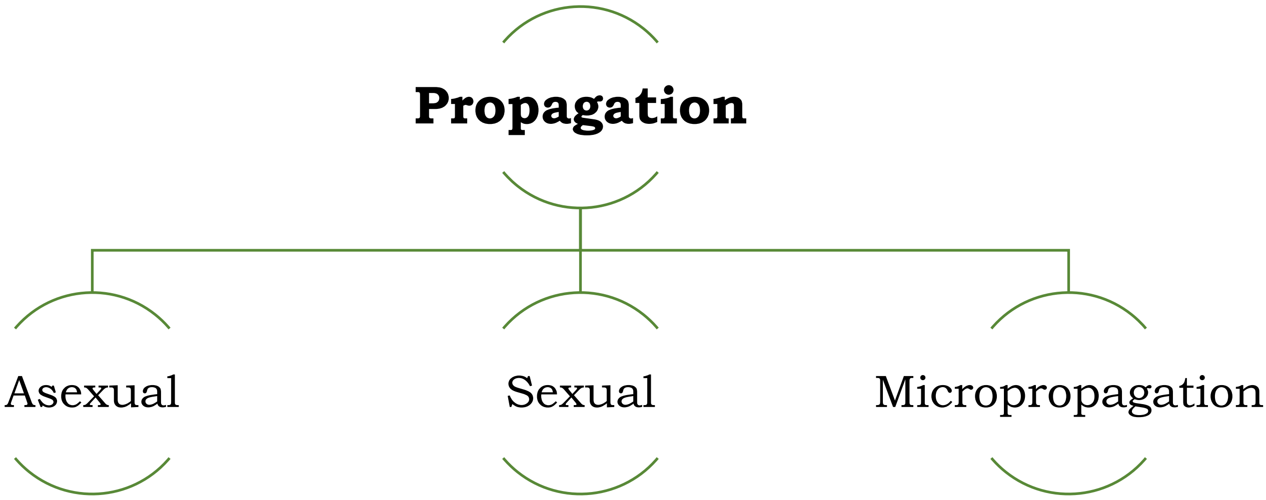
To improve **external skin colour & market acceptance, citrus fruits are treated with ethylene as a de-greening agent** because it breaks down the chlorophyll pigment of peel and yellow, orange colour to be expressed.

In India & some other countries, use **of calcium carbide for ripening process is banned**.

Plant Propagation, Methods of Propagation for Horticultural Crops

Plant Propagation **is defined as the multiplication of plants** by both sexual and asexual means. The horticultural plants are propagated both by sexual and asexual methods.

Most of the horticultural plants are now propagated through grafting and budding, few through cuttings, layering, seeds and micro-propagation. **The propagation methods are broadly classified as sexual, asexual and micro-propagation.**



Sexual Propagation

Sexual **propagation is the raising of plants by means of seed** which is formed due to the **fusion of male and female gametes** within the ovule of a flower. Plants that are produced from seeds are called seedlings

Example-Papaya, phalsa and mangosteen, **vegetable crops and flowers** are still being propagated by seed.

Asexual propagation

Propagation of plants **through any vegetative parts is called vegetative** or asexual propagation. The goal of vegetative **propagation is to reproduce progeny plants identical** in genotypes to a single source plant.

Methods of Asexual Propagation:

Asexual Plant Propagation

- By Cutting
- By Layering
- By Grafting
- By Budding

1. **Plant Propagation by Cutting-A portion of a stem, root or leaf is cut from the parent plant and is placed under certain favourable environmental conditions to form roots and shoots**. Thus, a new independent plant is produced which in most cases identical with the parent plant

a. Hard Wood Stem-Cuttings-

Propagation by hardwood cuttings is simple and cheapest method of multiplication. Hardwood cuttings are easily handled and transplanted. **One-year-old mature shoots are collected during November-February.**

- **Example** - Grape, fig, pomegranate, currant, gooseberry, some plums and apple are propagated by hardwood cuttings
- Cuttings should be **between 10 and 45cm.**
- It should contain at **least 2 buds**.

Figure 9 – Hard Wood Stem Cutting

- These cuttings are **treated with Naphthalene Acetic Acid (NAA)** and Indole-3-Butyric Acid (IBA) as they are the two most commonly used root promoting hormones.

b. Soft Wood Cuttings-

Cuttings are prepared **from the soft succulent new spring growth of species** which are **4 to 6 months old**.

- Many ornamental woody plants can be propagated by softwood cuttings.
- **Examples** - hybrid French lilacs, Foresythia, Magnolia, Spiraea, maples. Nerium, crotons, Eranthemum, Graftophyllum etc can also be multiplied through this type of cuttings.

Figure 10 - Soft Wood Cutting

c. Herbaceous Stem Cuttings:

Figure 11 - Herbaceous stem cutting

This type of cuttings is taken **from succulent herbaceous green house plants.**

Only 2-3 months old shoot and very tender shoots are utilized.

For example-Chrysanthemum, Coleus, Carnations, Geraniums, Cactus and many foliage plants are multiplied through herbaceous cuttings.

d. Leaf Cuttings:

Figure 12 – Leaf Cutting

Certain plants with thick and fleshy leaves have the capacity to produce plant lets on their leaf's

Example - bryophyllum, **Begonia**, African violets and peperomia are propagated by leaf cuttings.

e. **Root Cuttings**-

Figure 13 – Root Cuttings

Roots of the plant are utilized as propagating material. **Roots 1 cm thick and 10-15cm tall** are used.

- In temperate fruits, such kind of **roots are prepared in December** and kept in warm place in moss grass or wet sand for callusing **and transplanted during February-March in open** beds.
- **Blackberry and raspberry are commercially propagated** by this method. This method is also advocated in pecan nut, apple, pear and peach.

2. Propagation By Layering

Layering is the **method of propagation in which roots are developed on a stem while it is still attached to the parent plant**. **After proper rooting, the stem is detached and becomes a new plant for growing on its own roots.** The high success of layering is obtained by ringing or wounding, etiolation (absence of light), **use of rooting hormone (IBA,** NAA) **and favourable environmental condition (temperature and humidity**). The **layering can be natural means of propagation as in black raspberries** and trailing blackberries or can be artificially created by different means.

The Layering Techniques Generally Employed In Fruit Plants Are:

a. **Tip-layering**-:

In tip-layering, **rooting takes place near the tip of current season's shoot** which is **bent to the ground**.

- It is commonly **followed in black berries, raspberries and dewberries**.
- The stem of these plants completes their life in two years.
- The tips of these shoots are **buried 5-10 cm deep in soil**.
- Rooted layers are detached and planted in soil during spring.

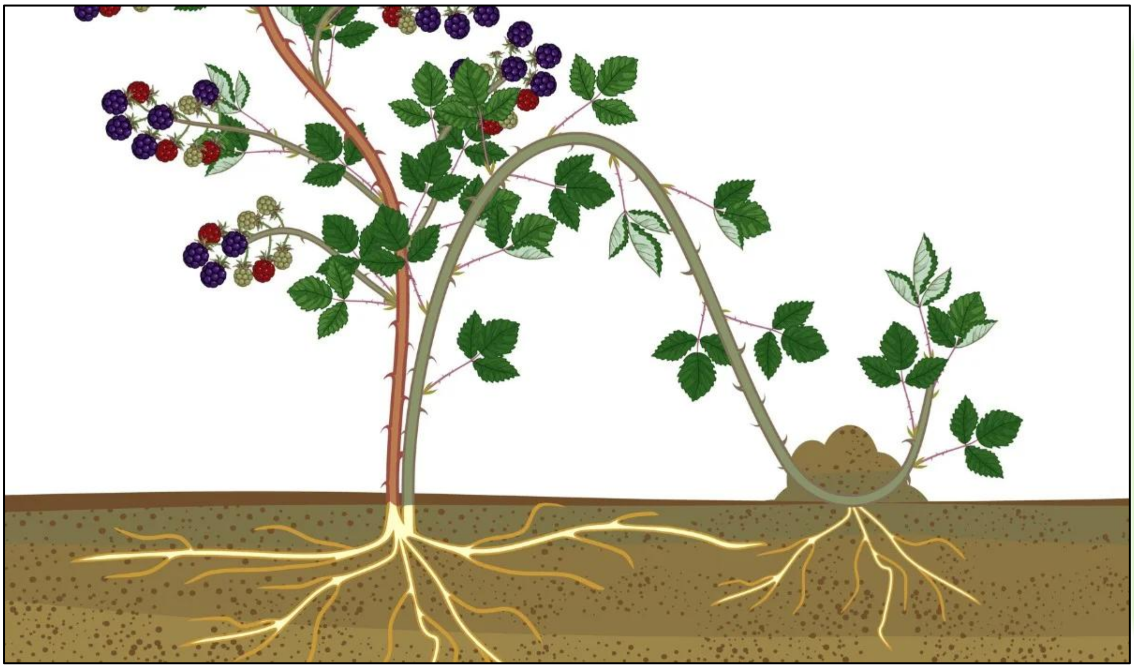


Figure 14

Tip Layering Of Blue Berries

b. **Serpentine layering**: It is modification of simple layering in which **one-year-old branch is alter natively covered and exposed.** The stem is girdled at its lower part. The **exposed part of stem should have at least one bud to develop a new shoot**. After rooting, the sections are cut and planted.

Example -Muscadine grape is commercially propagated by this method.

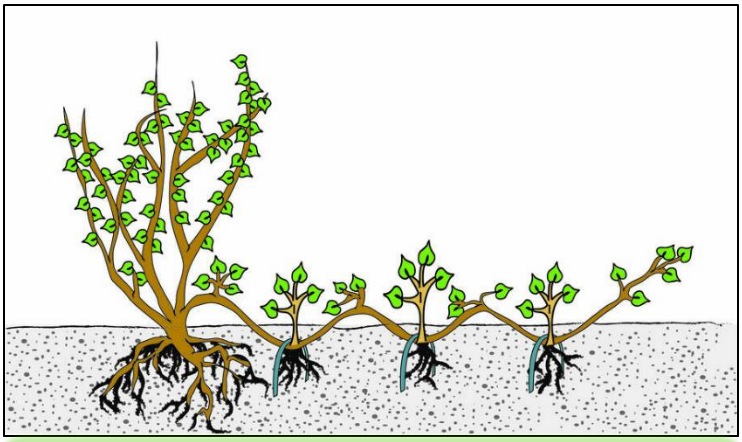


Figure 15 – Serpentine Layering

c. **Air-layering**:

In this method, **roots are formed in the aerial part of the plant**.

- The stem is girdled and rooting hormone (IBA) is applied to upper part of cut. The moist rooting medium (moss grass) is wrapped with the help of small polythene strip (200-300 gauze, transparent). **This method is commonly known as goottee**.

- **Example** - Many plants like litchi, kagzi lime, jackfruit, **guava and cashew-nut** as well as Ficus species, Croton, Monstera and philodendron are propagated through air-layering.
- **February-March and June-July are the ideal periods for air-layering**.
- Rooting in air layers generally commences **within 25-30 days** and layers are ready for **transplanting within 3 months.**

Figure 15 – Air Layering of Guava

d. <u>Mound layering/stooling</u>:

In this method, the plant is **headed back to 15cm above the ground level during dormant season.** The new sprouts will arise within 2 months.

- These sprouts are then **girdled near base and rooting hormone (IBA),** is applied to the upper portion of cut with moist soil.
- The rooting of shoots is observed **within 20-30 days**.
- **After 2 months**, the rooted shoots are separated from mother plants and planted in nursery.
- **Example**- Apple and pear root stocks and guava are commercially propagated by this method.

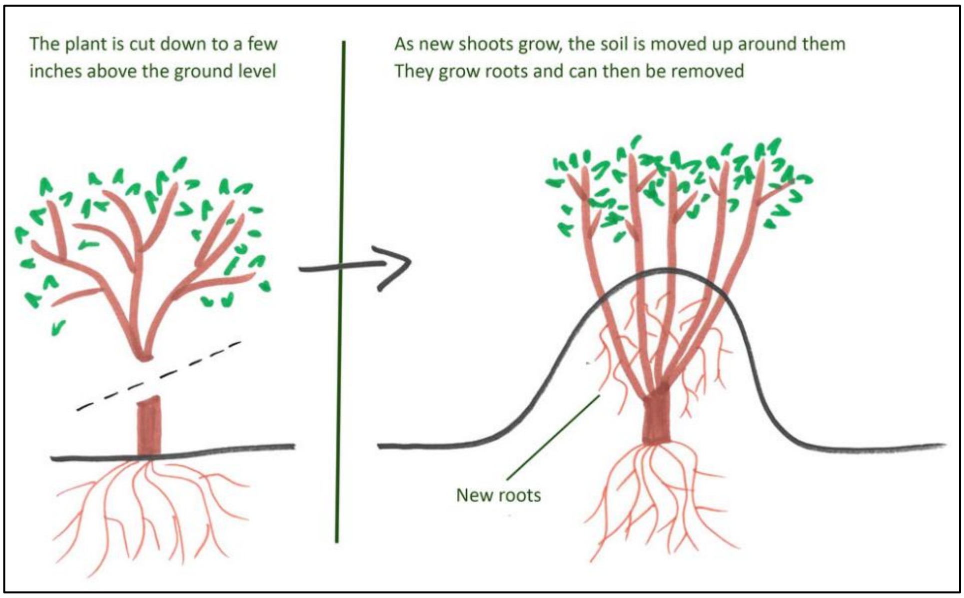


Figure 16 – Mound Layering

e. **Trench layering:**

Trench layering consists of growing a plant or branch of a plant in a horizontal position in the base of trench and filling in soil around the new shoots.

Roots are developed at the base of new shoots, so produced.

Example - **Rootstocks of apple, pear** and walnut are usually propagated by trench layering.

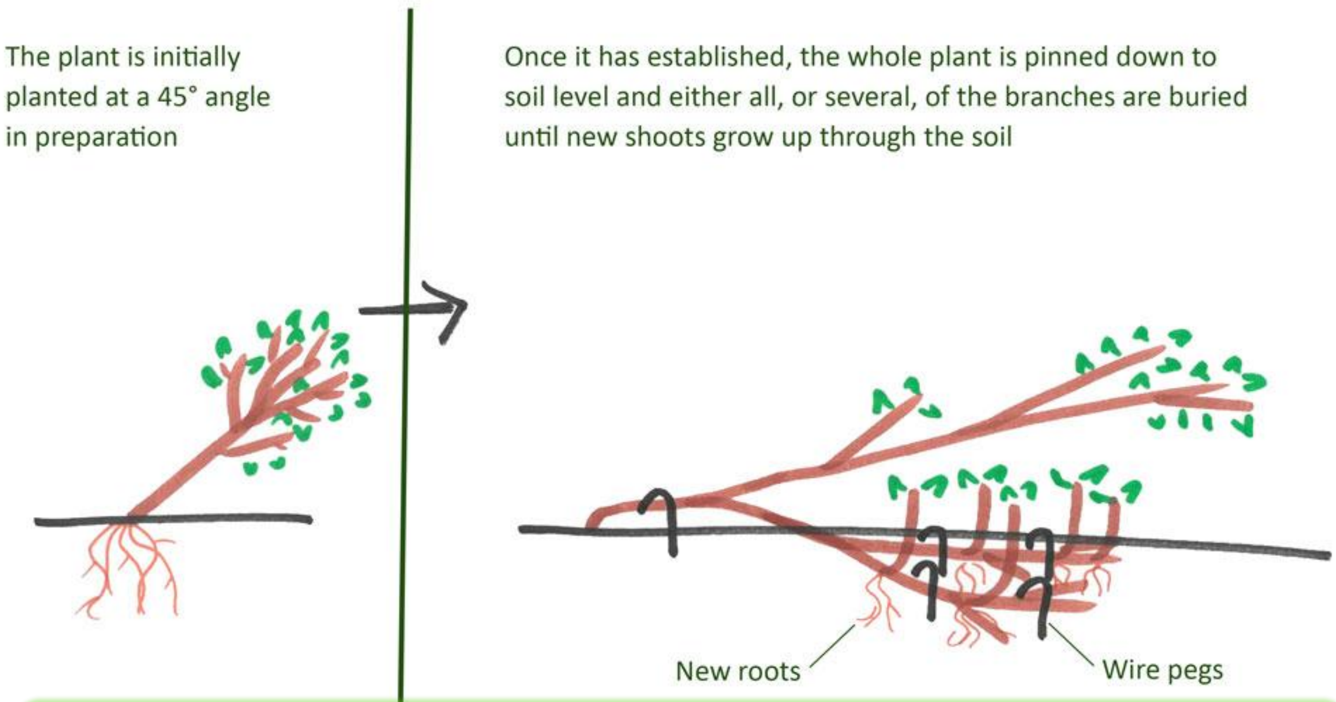


Figure 17 – Trench layering

3. Plant Propagation By Grafting

Grafting is an art of **joining parts of two independent plants in such a manner** that **they unite and grow together into single independent plant**.

- The part of graft combination which is to **become the upper portion** or the shoot system or top of the new plant is **termed as scion.**
- And the part which is to **become the lower portion or the root system** is the root stock **or under stock or some time stock**.

Methods Of Grafting

a. Inarching / Approach grafting:

The distinguishing feature of this method of grafting is that **two independent plants on their own roots (self-sustaining) are grafted together**.

- This method provides a means of establishing a successful **union between certain plants which are difficult to graft by any other method** as the two plants will be on their own roots till the formation of successful graft.
- **Examples -** Guava, mango, sapota

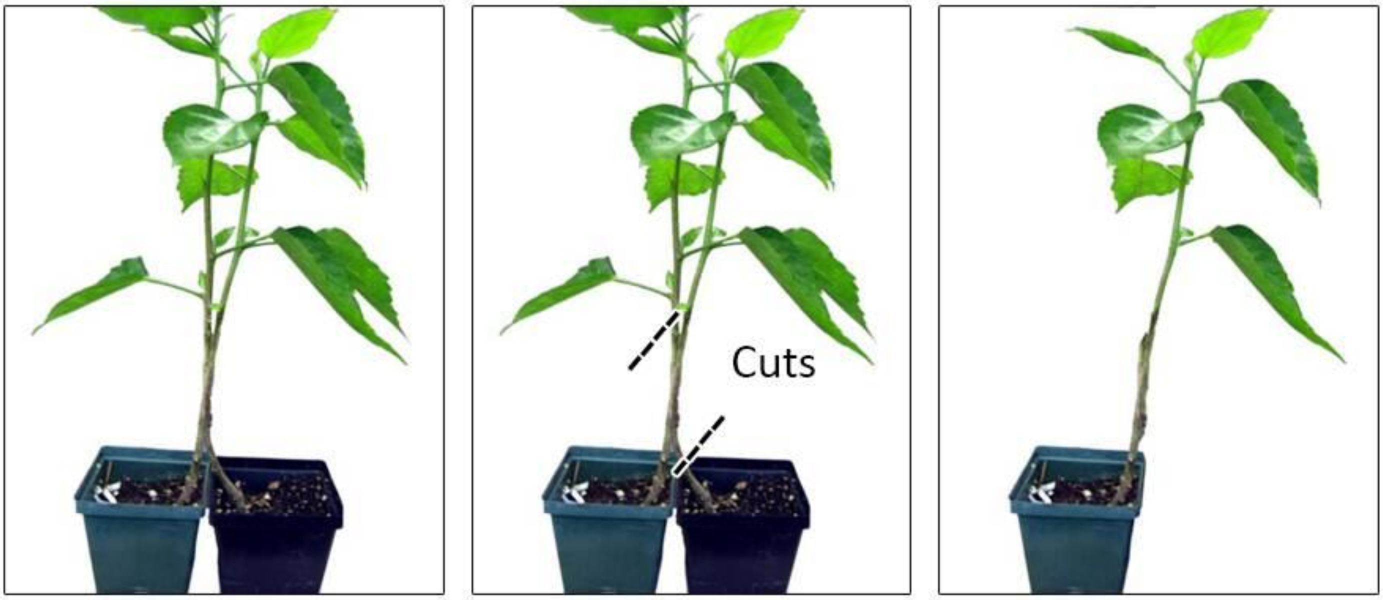


Figure 18 – Inarching Grafting in Mulberry (Two Rooted Plants Are Joined Together by Arching)

b. Veneer Grafting:

The method is simple and can be adopted with success. **Eight months to one year old seedlings are used as rootstock**s. In this method**, a downward and inward 3-4 cm long cut is made in the smooth area of the stock at a height of about 20 cm**.

- At the base of cut, a small shorter cut is given to intersect the first so as to remove the piece of wood and bark.
- **Proper selection** and preparation of scion are of utmost importance.
- The **scion stick is given a long slanting cut on one side and a small** short cut on the other so as to match the cuts of the rootstock.

- The **scion is inserted in the rootstock** and the graft union is then tied with polythene strip.
- The **rootstock should be clipped in stages when** the scion takes and remains green for more than 10 days.
- **Examples** - It is used widely for grafting plants such as Avocado, Mango etc.

Figure 19 - Veneer Grafting Mango

c. **<u>Tongue Grafting</u>**-

- In this method, **a flat slanting cut, about 5 cm long is given** at the **base of the scion** so that the lowest bud is about midway along the cut but on the opposite side.
- A **downward pointing tongue is made in the upper half** of the slanting surface.
- **A slanting cut, corresponding in length to that of the scion, is made upwards through the stock** 15-20 cm above the ground.
- An **upward pointing tongue** is made in the upper half of this slanting surface.
- The **cut surfaces of the scion and stock are now placed together so that the tongues interlock and the cambial regions are in close contact.**

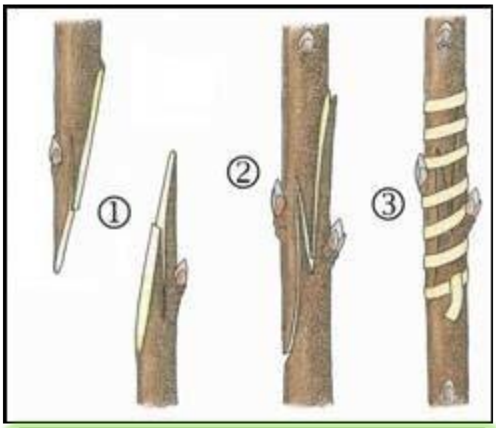


Figure 20 – Tongue Grafting

- **Examples -** apple, pear, peach, plum, apricot, almond, cherry, kiwi fruit, pecan nut etc.
- The best time for **tongue grafting is December- February i**n temperate fruits.

d. Epicotyl (Stone) Grafting

This method of grafting is done on the epicotyl region of the young seedlings; hence the name epicotyl grafting. This method is simple, **economical and useful for multiplication of mango plants in large number in a less time**. Fresh mango stones are sown in the nursery beds.

Figure 21 – Stone Grafting in Mango

Example- are Cashew, mango etc.

e. Cleft Grafting

The method involves **causing the graft union** by inserting the scion, which is in the form of a wedge, on to an incision on the stock plant. The scion is usually a branch excised from the mother plant. Sometime **2 or 3 scions** are inserted in stock or mother plants.

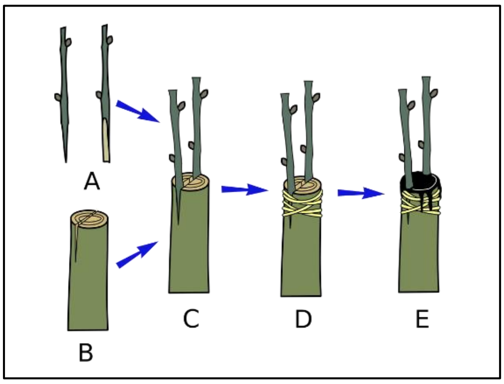


Figure 22 – Cleft Grafting

f. Softwood grafting

Grafting is done on the soft shoot of the stock plant which is a **seedling of about 6 months**.

Figure 23- Softwood Grafting

#Generally grafting is performed during dormant season whereas budding is done during active growing season

4. Plant Propagation by Budding

Budding is also a **method of grafting wherein only one bud with a piece of bark and with or without wood is used as the scion material.** It is also called as bud grafting. The plant that **grows after union of the stock and bud is known as budding**.

Methods of budding

a. T-Budding (Shield budding):

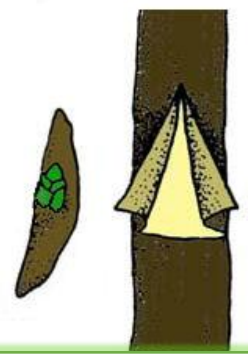
This method is **known as T-budding as the cuts given on the stock** are of the shape of the letter T, and shield budding as the bud piece like a shield. This method is widely used for propagating fruit trees and many ornamental plants.

Example -**Rose and Citrus**.

Figure 24 – T – Budding

b. Inverted T- Budding:

Practiced In heavy rainfall areas, water running down the stem of the stock may enter the T cut, soak under the bark and prevent healing of the bud piece. Under such conditions, an inverted T budding may give better results.



Inverted T budding procedure is same as that of **T budding except the horizontal cut on the stock is made** at the bottom of the vertical cut rather than at the top.

Figure 25 – Inverted T Budding

c. Patch Budding:

In this method a **regular patch of bark is completely removed from the stock plant** and is replaced with a **patch of bark of the same size containing a bud** from the desired mother plant.

- For this method to be successful the bark of the stock and bud stick should be easily slipping.
- **The diameter** of the stock and bud stick should **be preferably by about the same** (1.5 to 2.75cm).
- **Examples** are Pecan nut and walnut.

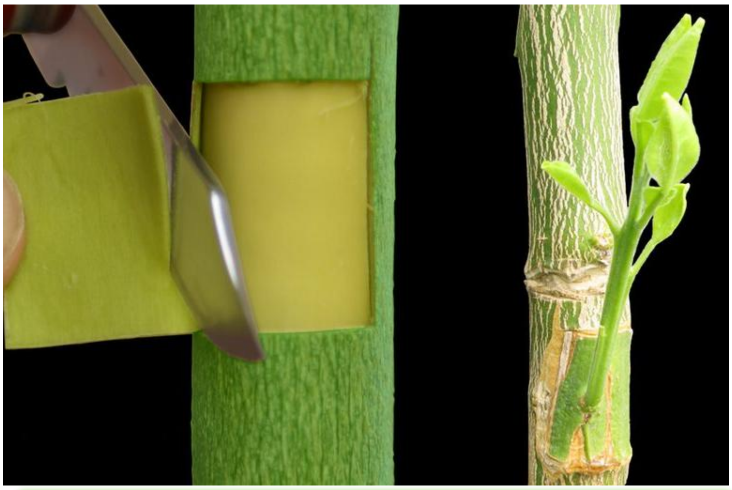


Figure 26 – Patch Budding

d. Ring budding:

The bud is prepared by taking a **ring of a bark, 3cm long with the bud in the centre**.

- In the root stock, two transverse cut 1.5cm apart are made and these are connected with a vertical cut and a ring of bark is removed.
- The **prepared scion bud with the ring of bark** is fitted in the exposed portion of the rootstock and tied.
- **Example is Ber**.

Figure 27 – Ring Budding

3. Micro-Propagation-

Micro propagation (tissue culture or invitro culture) refers to the multiplication of plants, in **aseptic condition** and in artificial growth medium from plant parts like meristem tip, callus, embryos anthers, axillary buds etc. It is a method by which a true to type and disease-free entire plant can be regenerated from a miniature piece of plant in aseptic condition in artificial growing medium rapidly throughout the year.

Apomixis

The phenomenon **in which an asexual reproductive process occurs in place of the normal sexual reproductive process of reduction division and fertilization is known as apomixis**. This is of great value as the resulting plant can be reproduced by seed propagation in almost the same manner as it would be by any other vegetative method. The seedlings produced through apomixes are known as apomictic seedlings.

Table 1 - Commercial Production Of Various Fruit Crops

S.No	Name Of Fruits	Scientific Name	Commercial Methods Of Propagation
1.	**Acid lime**	*Citrus aurantifolia.*	Seed
2.	**Aonla**	*Emblica Officinalis*	Patch budding
3.	**Bael**	*Aegle marmelos*	Patch budding
4.	**Ber**	*Ziziphus mauritiana*	Ring and T-budding
5.	**Grape**	*Vitis vinifera*	Hard wood stem cuttings
6.	**Grapefruit**	*Citrus × paradisi*	T-budding
7.	**Guava**	*Psidium guajava*	Stooling, Inarching
8.	**Jamun**	Syzygium cumini	Shield and Patch budding
9.	**Karonda**	*Carissa carandas*	Seeds, Air layering
10.	**Lemon**	*Citrus limon*	Air layering, hard wood stem cuttings
11.	**Litchi**	*Litchi chinensis*	Air layering
12.	**Mandarin**	*Citrus reticulata*	T/shield budding
13.	**Mango**	*Mangifera indica*	Inarching, Veneer grafting
14.	**Persimmon**	*Diospyros kaki*	Crown grafting
15.	**Phalsa**	*Grewia asiatica*	Seeds
16.	**Pomegranate**	*Punica granatum*	Hardwood stem cuttings and Air layering
17.	**Sweet orange**	*Citrus sinensis*	T-budding

Propagation By Specialized Vegetative Structures

Some fruit plants have natural structures-runner, sucker, offset, rhizome and crown-for propagation.

1. Runner:

It is a specialized stem **which is produced from the leaf axil at the crown of plant** and prostrate horizontally.

- The roots **appear at one of the nodes having contact with** soil.
- After root formation in the new plant, the contact with the mother plant **is automatically detached and new plant** can be separated and planted.
- **Strawberry is the typical example** which is commercially propagated through runners.

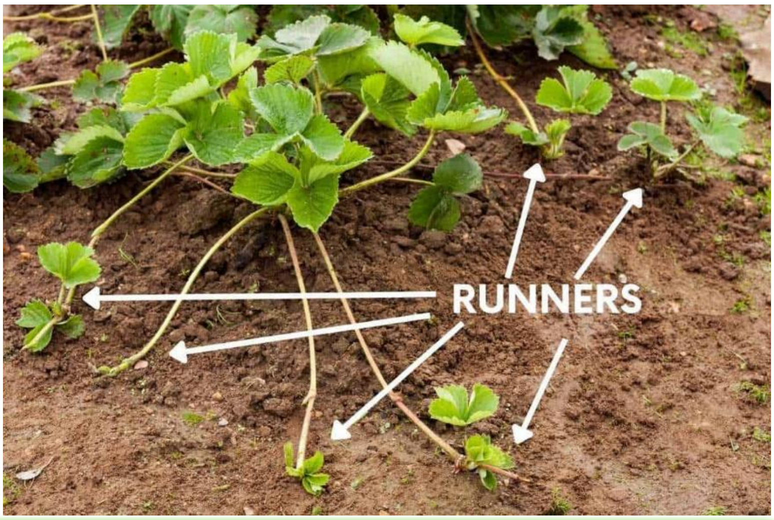


Figure 28 – Strawberry Propagation By Runners

2. <u>Suckers:</u>

A sucker is a shoot which arises on a plant below the ground. However, in practice, shoots **which arise from vicinity of the crown are also referred to as suckers**.

Figure 29 – Suckers Of Banana

Examples

- **Pineapple** is usually propagated through suckers.
- **Banana,** 2 types of suckers are produced-**water sucker and sword sucker**. Water suckers are broad leaved while sword suckers are pointed and in the shape of a sword.

3. Offset:

It is a **lateral shoot or branch** which is developed from base of the main stem.

Example -The **date palm and pineapple** produce such type of lateral shoots by which they can be propagated.

Figure 30 – Offset Of Pineapple

4. Rhizome:

Figure 31 – Rhizome of banana

A rhizome is a modified stem structure in which the main axis of the plant grows horizontally just below or on the surface of the ground.

Banana is a typical example where rhizome is cut into pieces in such a way so that each piece contains at least 3 lateral buds (eyes) for propagation.

5. Crown:

It **designates that part of a plant at the surface of ground from which new shoots are produced**. In **strawberry plant**, where leaves are seen in groups, is referred to as 'crown' of plant. Similarly, at the top is **the crown of pineapple plant**, which can be used for propagation purpose.

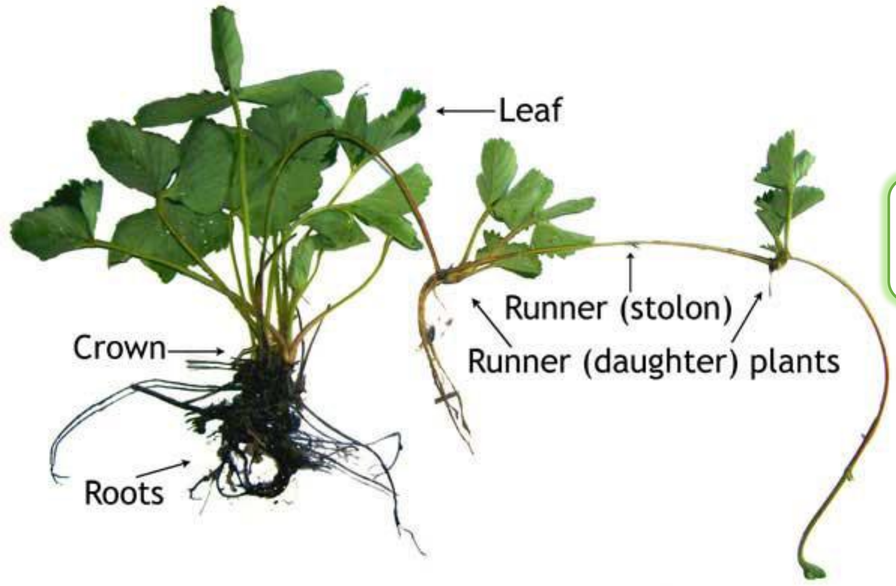


Figure 32 – Crown And Runner of Strawberry

6. **Corm:**

The **bulb consists predominantly of modified leaves; the corm is a modified stem**.

- Food is stored in this compact stem, which has nodes and very short internodes and is wrapped up in dry, scaly leaves.
- When a **corm sprouts into a new shoot**, the old corm becomes exhausted of its stored food and is destroyed as a new corm form above it.

Figure 33 – Corm Of Gladiolus

- The cormels **may be separated from the mother corm** at maturity (die back) and used to propagate new plants.
- **Examples** - **Amorphophallus, Colocasia, Gladiolus etc.**

7. **Stolon:**

It is a term used to describe various types of **horizontally growing stems that produce adventious roots when com**e in contact with the soil.

- These may be **prostate or sprawling stems** growing above ground.
- In propagating plants by stolon, the stolon can be **treated as a naturally occurring rooted layer and can be cut** from the parent plant and planted separately.
- **For example**, **Mint, Bermuda grass etc.**

Figure 34 – Mint Stolon

8. Stem tuber:

A tuber is **specialized swollen underground stem** which **possesses eyes** in regular order over the surface. The eyes represent the nodes of the tuber.

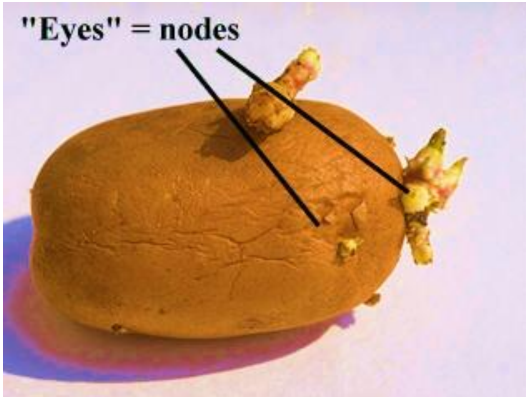


Figure 35 - Tuber Of Potato With Eyes

- The arrangement of the **nodes is spiral**, beginning with the terminal bud on the stolon to produce a new plant, **the tuber is divided into sections so that each section** has a good amount of stored food and a bud or eye.
- Propagation by **tubers can be done either by planting the tubers whole** or by cutting them into section, each containing a bud or eye. **For example, Potato**.

9. <u>Tuberous roots:</u>

These are **thickened tuberous growth** that functions as storage organs.

- These **differ from the true stem tuber**; in that they lack nodes and internodes. Buds are present only at the crown or stem end.
- **Fibrous roots are commonly** produced towards the opposite end.

Figure 36 – Tuberous Root of Dahlia

- **Most plants with fleshy roots** must be propagated by dividing the crown so that each section bears a shoot bud.
- **For example** -Dahlia, Begonia, Sweet potato

Propagation By Seed

Papaya, phalsa, kagzi lime and jamun are usually propagated by seeds. Seeds are also used to raise rootstock seedlings in many fruit crops such as citrus and mango.

Principles Of Horticulture Crop Production Technology

Principles Of Fruit Production

The commercial production of fruits is known as orcharding.

Grape plantations are called vineyards, and the cultivation of grapes is called viticulture.

Citrus orchards are typically called **citrus groves** and the cultivation of citrus is known as **citriculture.**

Orcharding and other types of fruit growing require **high capital investment** for years on a fixed site, without immediate return.

Selection Of Site

Selection of suitable site is the **first step** for establishing an orchard on commercial scale. Selection may be made based on the following criteria

- The land chosen for orchard **should be in proximity to main road and market**.
- It should have **proper irrigation facilities** and have a good soil and climate suitable for growth and production of fruit trees.
- **Experience of the fruit growers** and research stations in the locality should be taken into account for the acclimatization of the fruits under consideration.

Layout Systems of Planting

Lay-out means locating the position of trees, roads and buildings in the orchard being established, and systems of lay-out refer to the orderly ways of planting the trees.

It **is desirable** to have the trees planted in systematic way because Orchard operations like interculture and irrigation are carried out easily.

- It enables **equal distribution** of area under each tree.
- It results in **least wastage of land**.
- It **makes supervision easier** and more effective.
- There is room for systematic extension of the orchard.

There are **mainly five systems of planting** of fruit trees. In all these systems, trees are **planted in rows**.

A. Square system –

In **square system the trees are planted in four corners of a square keeping** the same distance between row to row and plant to plant in the same row. This is the most commonly followed system and is very easy to layout. **This system permits inter cropping and cultivation in two directions.**

B. Rectangular system-

In rectangular system the **trees are planted in the same way as in a square** system except that the **distance between rows to row will be more than the distance between plant to plant in the same row.** The **wider alley spaces available** between rows of trees permit easy intercultural operations and even the use of mechanical operations.

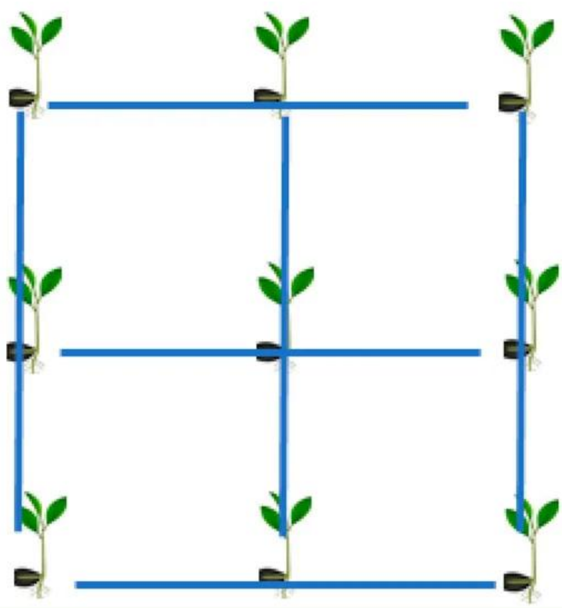


Figure 37 – Square System

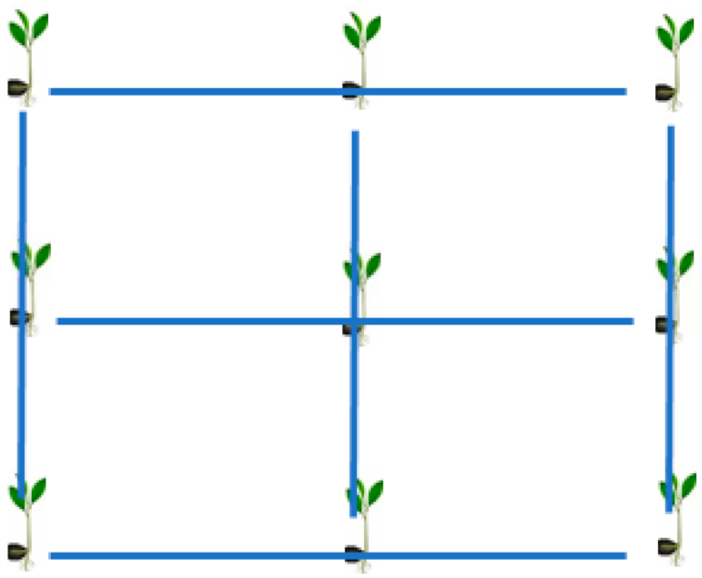


Figure 38 – Rectangular System

C. Quincunx Or Diagonal System –

Quincunx or diagonal system is the same as the square system except for the addition of a tree in the centre of each square.

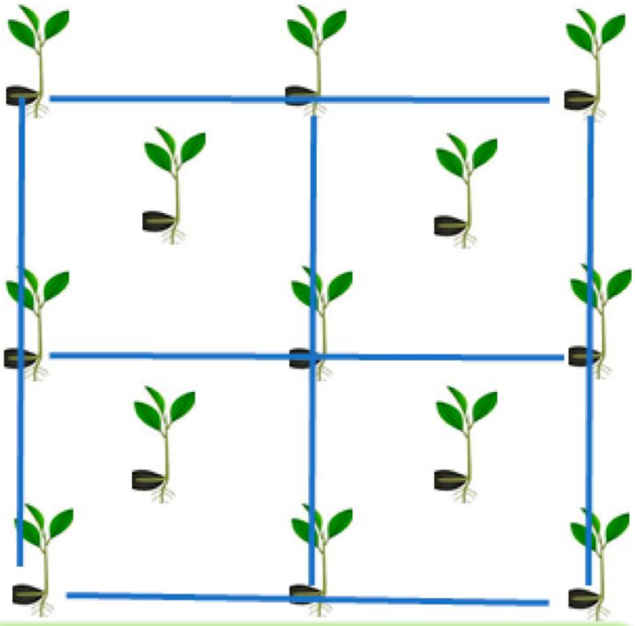


Figure 39 – Diagonal System

- This will accommodate double the number of plants, but **does not provide equal spacing**. The **central trees are known as filler crop** and the others as main crop.
- The central (filler) tree chosen may be a short lived one. Papaya, Guava, Lime, plum and peaches are a few examples of filler crops in orchards with trees like mango, jackfruit and tamarind.
- **This system can be followed when the distance between the permanent trees is more than 10 m**.
- As there will be **competition between permanent and filler** trees, the filler trees should be removed after a few years when main trees come to bearing.

D. Hexagonal/ septuple system in the hexagonal system,

The trees are **planted at the corners of an equilateral triangle**. **Six such triangles are joined together to form a hexagon**.

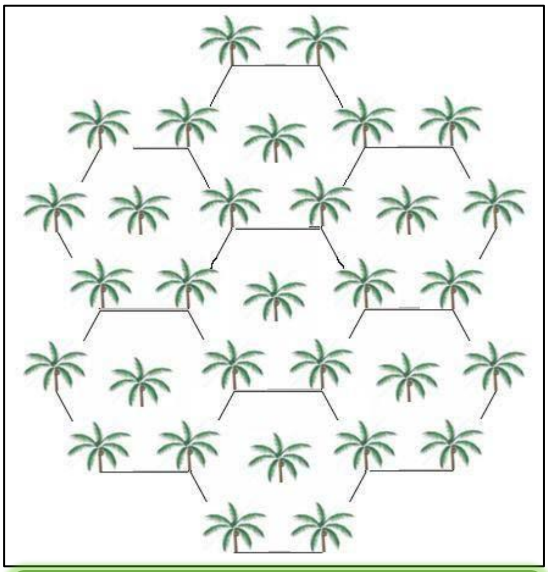


Figure 40 – Hexagonal System

- Six trees are positioned at the corners of this hexagon with a seventh in the centre all arranged in the three rows.
- This system provides equal spacing but it is difficult to layout.
- **This system accommodates 15% more trees than the square system**.

➢ **The limitations** of this system are that it is **difficult to layout** and the cultivation is not so easily done as in the square system.

E. Contour system

In a hilly area, a lot of depressions, ridges, furrows and place surface are found. But when **plating is done a line is made by connecting all the points of same elevation across the slope from a base line**. Thus, spacing is maintained on this row.

➢ The main purpose of this **system is to minimize land erosion and to conserve soil moisture** so as to make the slope fit for growing fruits and plantation crops.

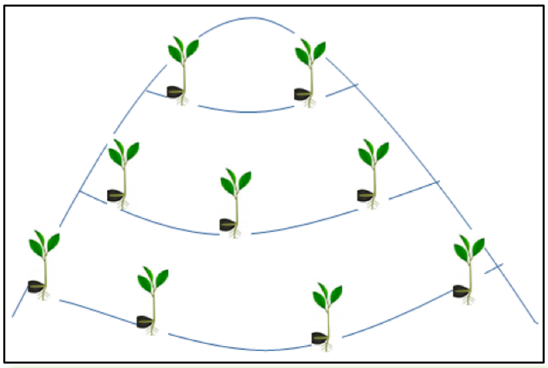


Figure 41 – Contour System

➢ **Terrace system on the other hand refers to planting in flat strip of land** formed across a sloping side of a hill, lying level along the contours.

In the recent past, a new trend has been **observed among the growers i.e. adoption of high-density planting system. This has gradually become imperative due to shortage of land and labour.**

High Density Planting System

High density planting (HDP) can be **defined as "accommodation of the maximum possible number of the plants per unit area to** get the **maximum possible profit** per unit of tree volume without impairing the soil fertility status"

Farm Operations Important for Horticulture Crops

Farm operations **like pruning and training of trees and thinning of fruits** are unique to **pomology a**nd are regular features of an orchard.

Training: It is a **physical technique that control the shape, size, and direction** of plant growth.

Pruning: It is a **judicious removal of plant part to improve shape**, influence growth, improve flowering, fruitfulness and fruit quality or to repair injuries.

Training

Physical techniques that control the shape, size and direction of plant growth are known as training or in other **words training in effect is orientation of plant in space** through techniques like **tying, fastening, staking, supporting over a trellis or pergola** in a certain fashion or pruning of some parts.

Objectives

- **To improve appearance and usefulness** of plant/tree through providing different shapes and securing balanced distribution.
- To **ease cultural practices** including inter-cultivation, plant protection and harvesting.
- To **improve performance like planting** at an angle of 45° and horizontal orientation of branches make them fruiting better.

Methods of Training

Method of training of a plant is **determined by the nature of plant, climate, purpose of growing, planting method, mechanization, etc**.

1. Training In Herbaceous Annuals And Biennials:

These plants are **usually grown without any attempt to alter their growth patterns** because even if useful not practical being in large number in field. However, for some of ornamental value and creeping nature following types of training is affected.

- **Staking or supporting** of vine like plants.
- **Training on pergola or trellis of vine type** fruit plants or even indeterminate type tomatoes.
- **Nipping of apices for encouraging** lateral growth to give bushy appearance or fulsome appearance in pot plants like aster, marigold and chrysanthemum.

Figure 42 – Training of Vine crop

- **De-shooting or removal of lateral buds for making single stem** for large flowers as in chrysanthemum and Dahlia.
- **Staking with bamboo sticks and tying together** various shoots in potted chrysanthemum.

2. Training Of Woody Perennials

The woody perennials, which are **widely spaced and remain on a place for a long duration**, are **trained for develop strong framework** for sustainable production of quality produce and for ornamental beauty in different shapes (topiary).

In these plants following types of training are followed.

1. **Open center system (Vase shaped)-** In this system th**e main stem is allowed to grow to a certain height and the leader is cut** to encourage lateral scaffold from near the ground giving a vase shaped plant. **This is common in peaches, apricots and ber**.

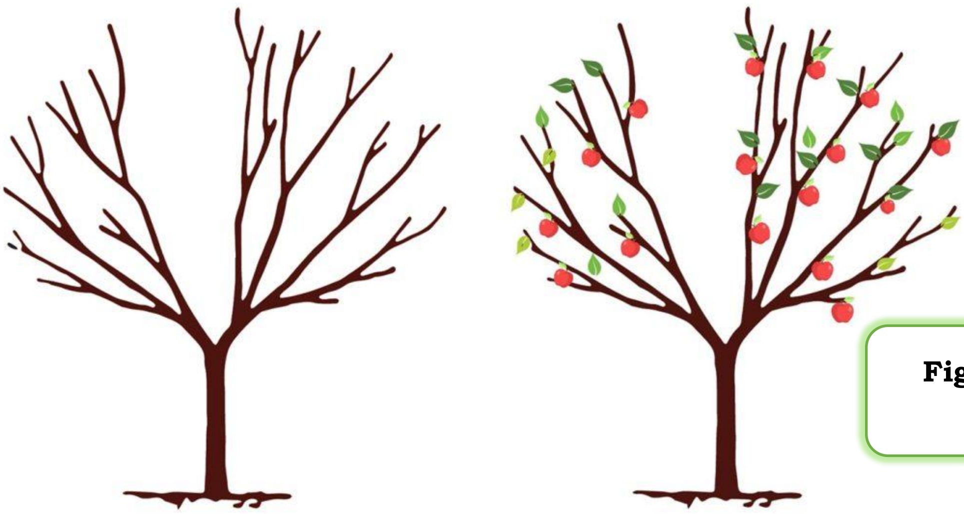


Figure 43 – Open center System

2. **Central leader system (closed centre)**:

- In this system the **central axis of plant is allowed to grow unhindered permitting branches** all around.
    - This system is also known as closed centre system and **common in use in apple, pear, mango and sapota**

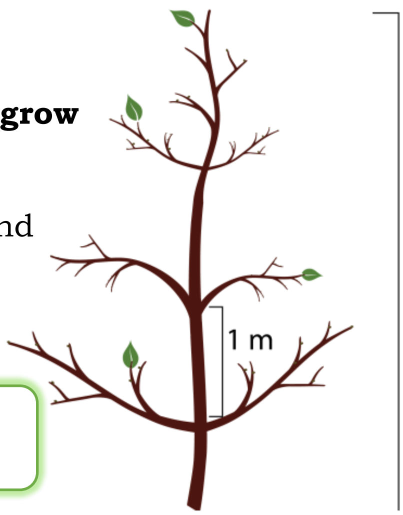
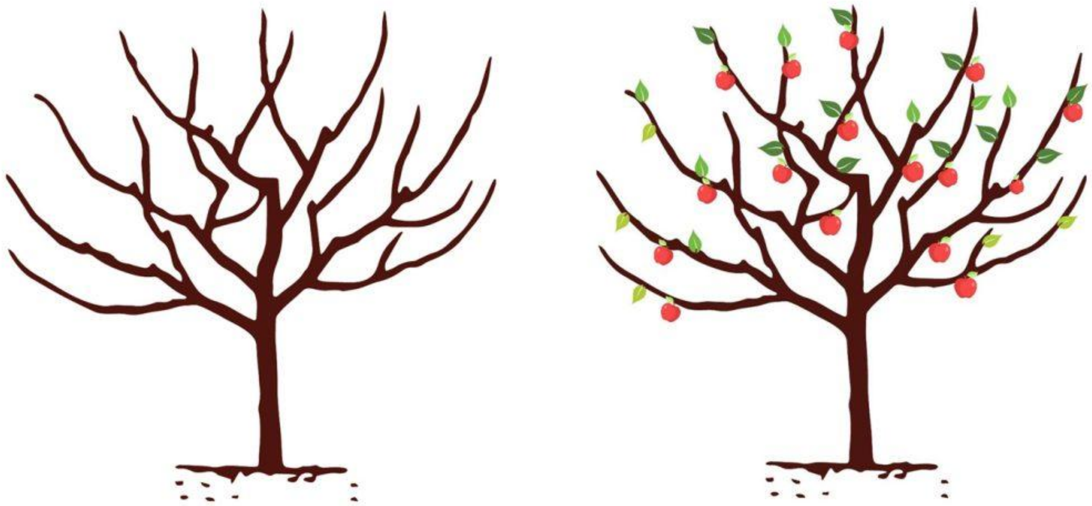


Figure 44 – Central Leader System

3. Modified leader system –

This system is in **between open centre and central leader system** wherein central axis is allowed to grow **unhindered up to 4—5 years and then the central stem is headed back and laterals are permitted.**



It is common in apple, pear, cherry, plum, guava

Figure 44 – Modified Leader System

4. <u>Cordon System/Trellis system</u>:

This is a system **wherein espalier is allowed with the help of training** on wires. This system is **followed in vines incapable of standing on their stem**. This can be trained in single cordon or double cordon and **commonly followed in crops like grape and passion fruit.**

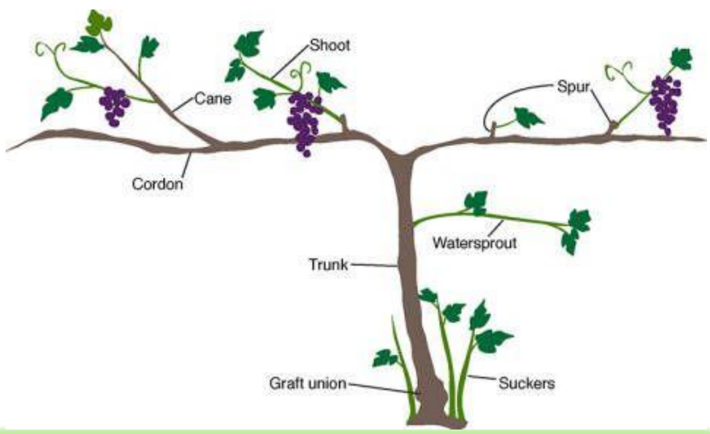


Figure 45 – Cordon System Of Training In Grape

5. <u>Training On Pergola</u>:

To support **perennial vine crops pergola is developed by a network of criss-cross** wires supported by RCC/angle iron poles on which vines are trained.

This is common for crops like grape, passion fruit, small gourd, pointed gourd and even peaches.

Figure 46 – Pergola System For Vegetable Crops

6. Training in different shapes:

Generally **ornamental bushes** are trained in different shapes for the purpose of **enhancing beauty of places**. **These shapes could be vase, cone, cylindrical and rectangular.**

Figure 47 – Training In Different Shape of Bushes

Pruning

It refers to removal of plant part like bud, shoot, root etc. to strike a balance between vegetative growth and production. This **may also be done to adjust fruit load** on the tree.

Objectives:

(i) To control plant size and form.

(ii) For plant performance like

- **Establishment of transplant where leaves/shoots are pruned** to strike a balance between root and shoot so that plants lose less water against restricted root system lost during lifting of plants.

- **Improvement in productivity** and quality by regulating the load of the crop and extent of flowering.
- **For flower and fruit quality**.
- **Elimination of non-productive vegetative** growth like water sprouts, suckers, dead and diseased wood.
- In case of forest trees **production of knot free timber**

Types Of Pruning:

Basically**, there are three types** of pruning with definite purposes.

1. **Frame pruning.**
2. **Maintenance pruning.**
3. **Renewal pruning.**

Frame Pruning:

This pruning is **done to provide shape and form to a plant in its formative years** so that **tree develops strong framework** and a shape for ease of operations. This process begins **from nursery itself and continues up to fruiting stage**. This is done continuously irrespective of the season

Maintenance Pruning:

To **maintain status- in production level and for uniform performance** this pruning is done. In some plants like **grapes, apple, pear, peach** etc. (deciduous trees) it is an annual feature and in others (evergreen like mango, sapota) it is rare confining to removal of water sprouts and unproductive growth and opening of the tree.

Renewal Pruning:

This pruning is **done in old trees like mangoes which shows decline**. In this case severe pruning is required.

Systems Of Prunings

1. **Heading back**: Only tops of branches are headed back or cut off (light pruning).
2. **Thinning out:** Complete removal of a branches or a part
3. **Dehorning**: Cutting away the main limbs or thick major branches

4. **Bulk pruning**: Heavy pruning all over the tree. For good fruit production only judicious heading back or thinning out should be done.

Factors To Be Considered In Pruning

In some of the tree species pruning as a regular feature in bearing **trees is done to strike a balance between vegetative growth and production** so that farmers get sustained production uniformly with optimum quality of produce. To achieve this one should consider the following factors.

(i) **Time at which buds are differentiated** in relation to blooming.

(ii) **The age of the wood that** produces the most abundant and highest quality of fruit buds.

Special Horticultural Practices For Inducing Fruiting

In **vegetative weak plants, favourable conditions for carbohydrate accumulation** will have to be created by provision of desired temperature, light, water and nutrient change while in vigorous plants this change is to be induced by the reduction of water and nutrient supply. **Regulation of fruiting can be affected by influencing fruit bud differentiation or by influencing fruit set and development.**

Pruning, root pruning, ringing, girdling, notching, bending, smudging are some of the specialized horticultural practices followed for regulation of fruiting.

1. Ringing and Girdling

Ringing consists of **removing a ring of bark about 1 to 2 cm wide** around the trunk or branches, **while Girdling is a milder treatment to draw a knife around the branch** so as to cut through the bark but not

Figure 48 - A. Girdling and B. Ringing

the wood. A wire tied very firmly round the stem also serves the same purpose.

- Ringing **or girdling will increase the concentration of carbohydrates** above the wing.
- Ringing is a **drastic operation** done when fruit **trees fail set fruit**.
- It is likely to **check vegetative growth** and to some extent the growth of roots.
- Ringing is done in **vigorous mango tree**.

2. Notching

It is similar to ringing except that **in notching only soil slip bark about 0.2 to 0.5 cm thick and 1.5 to 2.5 cm in length is removed just above or close to a dormant bud** in slantwise so that the latex does not coagulate in the bud itself.

Figure 49 - Notching

- Generally, **3 to 4 buds in the middle portion** of the selected shoot are best to operate on. Fig has responded to notching and it is practiced in fig cultivation.
- The season for notching the **fig is August-September**.

3. Bending

Bending a branch downward, sometimes checks growth and causes accumulation of starch in the **branch with greater flowering. This tends to increase carbohydrate concentration.**

Figure 50 – Bending Of Guava Stems

- The bending of **branches is usual as a substitute for severe pruning** in shaping the young trees and more fruit is borne because more branches are left to bear and more leaves are left to synthesize food material.

- In case of **bending the effect of apical dominance** of the growing shoot is removed and auxins during translocation activate the dormant buds.
- This is usually **practiced with local guava variety in the Maharashtra** state (Deccan area).

4. Smudging

Smudging is a **practice of smoking the tree by burning brush wood on the ground and allowing smoke to pass through the centre of the crown of the tree**.

- The smoking is discontinued as soon as the terminal buds begin to swell.
- Practice of **smoking to the trees like mango, commonly employed in the Philippines to produce off-season crop**.
- Smoking **containing ethylene gas**, which is responsible for initiation of flowering.

5. Root pruning

Root pruning **results in less carbohydrate utilization of the top growth** through there is a little more utilization of carbohydrate for root functions. There is an accumulation of carbohydrates due to check of top growth, which results in fruit bud differentiation.

- As the effect of **root pruning is to check the vegetative growth**. The plant became dwarf. **Root pruning is a method of inducing fruitfulness** or determining the time of flowering. The root pruning is done two months before the bloom required.
- **The main roots are exposed to the sun and the fibrous roots are cut**, so water is withheld. The trees are allowed to go dry until their leaves wither and fall down. The time taken for leaf fall is from 3-4 weeks.
- After that **exposed roots are covered with a mixture of soil and** manure. The trees are then immediately irrigated.
- The trees burst into flowering in about 2-3 weeks. **Practice very widely adopted by citrus growers in western and central India (in santra)**.
- The trees on which root pruning is practiced quite frequently are short lived and are liable to be week and unhealthy. Hence root pruning is usually restored to when other method such as ringing etc.

- **Root pruning** is generally included in bahar treatments given to fruit **trees like mosambi, santra, guava, pomegranate, lime etc.**

6. Bahar treatment

This practice is followed with fruit trees like mosambi, santra, grape fruit, guava, pomegranate ber, lime etc. in the state of Maharashtra, M.P. and Gujarat etc.

As there is no distinct winter (very cold winter) these fruit trees are usually continuous vegetative growth resulting in indistinct flowering season.

This practice is useful **in encouraging flowering as well as regulating the time**. About 1 to 1 ½ months prior to the expected flowering irrigation is withholding.

There are three flowering seasons namely Mrig bahar, Hasta bahar and Ambe bahar.

- **Mrig bahar**: Flowering in **June-July**
- **Hasta bahar**: Flowering in **October-November**
- **Ambe bahar**: flowering in **December-February**

The orchard is **ploughed up to 20 cm depth both ways and the roots are exposed by removing the upper 10-15 cm of soil within a radius of 60-90 cm around the trunk**.

- The dead and decayed fibrous roots are removed in the area exposed. The leaves start turning yellow, shrivel and fall. These are the indication to know that the trees have rested long enough and accumulated food reserves.
- The **exposed roots are then recovered with original soil** and necessary manures are added. Trees are irrigated lightly.
- The **second watering is given on the 3rd or 5th day and first two** watering stimulate blossoming and if heavy irrigations are given at the beginning, this may tend to vegetative growth only.
- **Root exposure is not necessary in case of sandy**, sandy loam and other types of light soils. The **choice of bahar depends upon availability of water** and time of year the fruit is required in the market.
- Where irrigation water is available, the grower prefers Hasta or Ambe bahar.

Growth Regulators

Growth retardants **like CCC, (in Mango 1000 ppm.),** higher **concentration of Auxin**, **NAA (in Mango 200 ppm**), **Ethylene** -750 ppm, Paclobutrazol -4-6 g (Cultar).

Use of hormones –

It is done in following activities:

- Rooting of cutting
- Blossom thinning
- Preventing fruit drop
- Increasing fruit setting or development of parthenocarpy fruits
- Germination of seeds
- Early maturity
- Weed control

Generally, Auxin, Gibberlin And Etylene Are Used In Horticulture For Rooting And Fruit Production

Harvesting, Handling and Storage of Horticultural Crops

All fresh **horticultural crops are perishable in nature** due to high in water content and are subjected to desiccation (wilting, shriveling) and to mechanical injury. Various authorities **have estimated that 20-30 percent of fresh horticultural produce is lost after harvest and** these losses can assume considerable economic and social importance.

These perishable commodities need very careful handling at every stage so that deterioration of produce is restricted as much as possible during the period between harvest and consumption. Postharvest rots are more prevalent in fruits and vegetables that are bruised or otherwise damaged. **Mechanical damage also increases moisture loss. The rate of moisture loss may be i**ncreased by as much as 400% by a single bad bruise on an apple, and skinned potatoes may lose three to four times as much weight as non-skinned potatoes.

Damage can be prevented by training harvest labour

- **To handle the crop gently**;
- Harvesting at proper maturity;

- **Harvesting dry** whenever possible;
- **Handling each fruit** or vegetable no more than necessary (field pack if possible);
- **Installing padding** inside bulk bins;
- And avoiding over packing of containers.

Harvesting Of Fruits And Vegetables

Quality cannot be improved after harvest, only maintained; therefore, it is important to harvest fruits and vegetables at the proper stage and size and at peak quality.

- Immature or over-mature produce may not last as long in storage as that **picked at proper maturity.**
- Harvest should be **completed during the coolest time of the day, which is usually in the early morning,** and **produce should be kept shaded in the field**.
- Fruits harvested too early may lack flavour and may not ripe properly, while produce harvested too late may be fibrous or have very limited market life.

Fruits should be harvested at the proper maturity stage. The maturity has been divided into two categories i.e. physiological maturity and horticultural maturity.

Physiological Maturity: It is the stage when a **fruit is capable of further development** or ripening when it is harvested i.e. ready for eating or processing.

Horticultural Maturity: It refers to the stage of **development when plant and plant part possesses the pre-requisites for use by consumers** for a particular purpose i.e. ready for harvest.

Maturity Indices Of Fruits And Vegetables

Post-harvest physiologists distinguish **three stages in the life span of fruits and vegetables: maturation, ripening, and senescence.**

1. **Maturation** is indicative of the fruit being ready for harvest. **At this point, the edible part of the fruit or vegetable is fully developed in size,** although it may not be ready for immediate consumption.
2. **Ripening** follows or overlaps maturation, **rendering the produce edible**, as indicated by taste.

3. **Senescence** is the last stage, characterized **by natural degradation of the fruit** or vegetable, as in loss of texture, flavour, etc. (senescence ends at the death of the tissue of the fruit).

The importance of maturity indices are to ensure sensory quality (flavour, colour, aroma, texture) and nutritional quality, to ensure an adequate post-harvest shelf life, to facilitate scheduling of harvest and packing operations.

Types Of Maturity Indices

1. **Visual appearance**:

The visual appearance of fruit and vegetable is the most important quality factor, which decides its price in the market. **Medium size produce is always preferred by the consumers,** because they tend to view large fruits as more mature.

a. **Size and shape**: Maturity of fruits can be assessed by their final shape and size at the time of harvest. Fruit shape may be used in some instances to decide maturity.

b. **Colour**: **The loss of green colour of many fruits is a valuable guide to maturity**, a complete loss of green colour with the development of yellow, red or purple pigments

2. **Physical indices**

a. **Firmness:**

As fruit **mature and ripen** they **soften by dissolution of the middle lamella of the cell walls**. The degree of firmness can be **estimated subjectively by finger or thumb pressure**, but more precise objective measurement is possible with **pressure tester or penetrometer**.

Figure 51 - Penetrometer

In many fruits such as **apple, pear, peach, plum, guava, kinnow** etc. firmness can be used to determine harvest maturity.

b. **Specific Gravity**:

As fruit mature, **their specific gravity increases. This parameter is rarely used in practice** to determine when to harvest a crop. **To do this the fruit or vegetable is placed in a tank of water; those that float will be less mature that those that sink.**

Figure 52 – Specific Gravity Test Of Grapes

3. Chemical Measurement

- The conversion **of starch to sugars during maturation is a simple test** for the maturity of some apple cultivars. It **is based on the reaction between starch and iodine to produce a blue or purple colour.**
- **The total soluble solids of** the fruit can be measured with refractometer, which indicate the harvest maturity of fruits.
- **Acidity is readily determined** on a sample of extracted juice by titration with 0.1 N NaOH.

4. Calculated Indices

a. Calendar date- For **perennial fruit crops** grown in seasonal climate which are more or **less uniform from year to year, calendar date for harvest** is a reliable guide to commercial maturity

- **Heat Units**: An objective **measure of the time required** for the development of the fruit to maturity after flowering can be **made by measuring the degree days or heat units in a particular environment.**
- In warm condition maturity will be advanced and under cooler conditions, maturity is delayed.
- The average or characteristic **number of degree-days is then used to forecast the probable date of maturity** for the current year and as maturity approaches, it can be checked by other means.

Table 2 - Maturity Index Of Various Fruits

Fruits/ Vegetables Maturity indices or characteristics	**Fruits/ Vegetables Maturity indices or characteristics**
Apple	12% SSC, 18 lb firmness
Banana	Disappearance of angularity in a cross section of the finger
Grapes	Minimum TSS 14 to 17.5%, depending on cultivars, TSS/TA ratio 20 or higher
Guava	Colour break stage (when skin Colour changes from dark green to light green)
Kinnow	TSS/acid ratio 12:1 to 14:1
Mango	Changes in shape (increase fullness of cheeks or bulge of shoulder), flesh Colour yellow to yellowish-orange
Brinjal	Brinjal Immature, glossy skin, 40days from flowering.
Carrot	Carrot Immature, roots reached adequate size.
Potatoes	Potatoes Harvest before vines die completely, cure to heal surface wounds.
Tomatoes	Seeds fully developed; gel formation advanced in at least one locule.
Watermelon	Watermelon Flesh Colour 75% red, TSS = 10%

Harvesting Of Fruits And Vegetables

Fruits and vegetables are living entities and therefore, should be harvested with great care. **Harvesting practices should not cause a little mechanical damage** to produce as possible.

The Following Points Should Be Kept In Mind While Harvesting The Crop:

- **Gentle picking and** harvesting will help reduce crop losses.
- **Wearing cotton gloves, trimming finger nails**, and removing jewellery such as rings and bracelets can help reduce mechanical damage during harvest.
- Produce should be **harvested during coolest part** of the day not wet from dew or rain.

- Empty **picking containers** with care.
- Keep produce cool after harvest (provide shade)

Method Of Harvesting Of Fruits And Vegetables

Harvesting of crops can be done **manually or mechanically**.

1. **Hand harvesting**: Usually done for fruits **destined for fresh markets**. The primary advantage is harvesting of fruit or vegetable can be done at appropriate maturity and produce will suffer minimum damage. However, it is a time-consuming process and more labour is required during harvesting season.
2. **Mechanical harvesting**: **Harvesting is carried out with the help of machines**. **The primary advantage** is that the produce can be **harvested at a faster** rate and less manpower is required as compared to hand harvesting. However**, damage can occur to crops** and this method is not much suitable for marketing of fresh commodities.

Postharvest Handling

Post harvest handling is the stage of crop **production immediately following harvest, including cooling, cleaning, sorting and packing.** The instant a crop is removed from the ground, or separated from its parent plant, it begins to deteriorate. Postharvest treatment **largely determines final quality**, whether a crop is sold for fresh consumption, or used as an ingredient in a processed food product.

The most important goals of post-harvest handling are

- Keeping the **product cool**,
- To **avoid moisture loss** and slow down undesirable chemical changes,
- **Avoiding physical damage** such as bruising, to delay spoilage

Handling Operation After Harvest

1. **Dumping: The first step of handling is known as dumping**. It should be done gently either using water or dry dumping. **Wet dumping can be done by immersing the produce in water**. It reduces mechanical injury, bruising, abrasions on the fruits, since water is gentle on produce. **The dry dumping is done by soft brushes fitted on the sloped ramp or moving conveyor belts. It will help in removing dust and dirt on the fruits.**

2. **Pre-sorting**: It is done to **remove injured**, **decayed, mis-shapen fruits**. It will save energy and money because culls will not be handled, cooled, packed or transported. **Removing decaying fruits** are especially important**, because these will limit the spread of infection to other healthy fruits during handling.**

3. **Washing and cleaning**: Washing with **chlorine solution (100-150 ppm)** can also be used to control inoculum build up during pack house operations. For best results, **the pH of wash solution should be 6.5-7.5. Mangoes, bananas** should be washed to remove latex.
4. **Grading: Grading can be done manually or by automatic grading** lines. Size grading can be done subjectively with the use of standard size gauges. The grading of fruits plays an important role in domestic and export marketing of fruits. Different fruits have different grades on the basis of their size and weight. **The grades of some fruits and vegetables decided by Directorate of Marketing and Inspection (DMI)**

Pre-cooling of horticulture produce

Pre-cooling of the produce soon after their harvest is one of the important components of the cool chain, which ultimately affect the shelf life of the produce. **The main purpose of precooling is to immediately remove the field heat from the produce.**

1. **Room cooling**: It is **low cost and slow method** of cooling. In this method, produce is simply loaded into a cool room and cool air is allowed to circulate among the cartons, sacks, bins or bulk load.
2. **Forced-air cooling**: Forced air-cooling is mostly used for wide range of horticultural produce. This is the fastest method of pre-cooling. **Forced air-cooling pulls or pushes air through the vents/holes in storage containers**. In this **method uniform cooling** of the produce can be achieved if the stacks of pallet bins are properly aligned.
3. **Hydrocooling**: **The use of cold water is an old and effective cooling method used for quickly cooling a wide range of fruits and vegetables before packaging**. This method of cooling not only avoids water loss but may even add water to the commodity. **The hydrocooler normally used are of two types i.e., shower type and immersion type**

4. **Vacuum cooling**: Vacuum cooling take **place by water evaporation from the product at very low air pressure.** In this method, **air is pumped out from a larger steel chamber in which the produce is loaded for pre-cooling**. Vacuum cooling causes **about 1 per cent produce weight loss** (mostly water) for each 6oC of cooling.
5. **Package-icing**: In some commodities, **crushed or flaked ice is packed along with produce for fast cooling.** The ice keeps a high relative humidity around the product. Package ice may be finely crushed ice, flake ice or slurry of ice**. Packaged icing can be used only with water tolerant, non-chilling sensitive products** and with water tolerant packages (waxed fibreboard, plastic or wood).

Packaging Of Horticulture Produce

Packing is a coordinated system of **preparing goods for transport, distribution, storage, retailing and end use and a means of ensuring safe delivery to the ultimate consumer in sound conditions at minimum cost**. It **protects the produce from pilferage**, microorganisms and adverse weather condition and it is also used to advertise the product.

Requirements Of A Good Package

- Should be **environment friendly**.
- Should have **sufficient strength** in compression and against impact and vibrations should be stable during the entire distribution chain.
- Should be **compatible with the automatic packing/filling**, handling machines (mechanical filling systems)
- Should **facilitate special treatments like** pre-cooling.
- Should have **consumer appeal**.
- Should be **easily printable**.
- should be **cost effective**.

Materials For Packaging

- **Wood** - boxes, bins, trays, barrels, pallets
- **Jute**/canvas - sacks
- **Paper and card board** - liners, boxes, trays

- **Plastic** - Rigid - crates, pallets, trays, Flexible - films (single & multi layered), Polystyrene boxes / trays
- **Combined materials** - CFB and plastic

Storage Of Fruits and Vegetables

The management of **temperature and relative humidity are the most important factors** determining storage life of horticultural produce. **The natural means like ice, cold water, night temperature** have been used for long time for protecting food materials from spoilage and these are still common. However, **with the development of innovative technologies, it is possible to achieve optimal environments in the insulated stores**.

1. Cold Storage:

Lowering the temperature to the lowest safe handling temperature is of paramount importance for **enhancing the shelf life**, reducing the losses and maintaining higher quality during marketing. **Damaged produce will lose water faster** and have higher decay rates in storage when compared to undamaged produce

2. Controlled Atmosphere (CA) Storage

The term implies the **addition or removal of gases resulting** in an atmospheric composition different from that of normal air. **Thus, the levels of carbon dioxide, oxygen, nitrogen, ethylene, and metabolic volatiles in the atmosphere may be** manipulated. **Controlled atmosphere storage generally refers to keeping produce at decreased oxygen and increased carbon dioxide concentrations and at suitable range of temperature and RH.** Systems where atmospheric control is accurately controlled are generally called CA storage

Modified Atmospheric Packaging

MAP (modified atmospheric packaging**) produce is enclosed in polymeric films** and is **allowed to generate its own atmosphere** (passive MAP) or air of known composition is flushed into the bag (active MAP) and depending upon gas / vapour transmission characteristics of the film on appropriate atmosphere develops in the package to prolong shelf life. **MAP is ideally combined with temperature control for maximum benefit**.

Benefits of CA storage

- **Slow down respiration and ethylene production rates**, softening and retard senescence of horticultural produce.
- **Reduce fruit sensitivity** to ethylene action
- **Alleviate certain physiological disorders** such as chilling injury of various commodities, russet spotting in lettuce, and some storage disorders including, scald of apples.

Harmful Effects of CA Storage

- **Initiation or aggravation of certain physiological disorders** can occur, such as blackheart in potatoes, brown stain on lettuce, and brown heart in apples and pears.
- **Irregular ripening of fruits,** such as banana, mango, pear and tomato, can result from exposure to O2 levels below 2% or CO2 levels above 5% for more than 2 to 4 weeks.
- **Off- Flavors and off-odours** at very low O2 or very high CO2 concentration may develop as a result of anaerobic respiration and fermentative metabolism.